Derivation of the CCA solution

For the first canonical variate pair, we seek linear combinations $U = a^\top X$ and $V = b^\top Y$ that maximize their correlation ρ subject to unit-variance constraints.

Equivalently,

$$\max_{a,b} \quad a^\top \Sigma_{12}\, b \quad \text{s.t.} \quad a^\top \Sigma_{11}\, a = 1, \quad b^\top \Sigma_{22}\, b = 1.$$

1. Set up Lagrange equations

As we have a continuous objective function with equality constraints, we can use the method of Lagrange multipliers. Accordingly, we introduce Lagrange multipliers λ, μ via

$$L(a, b, \lambda, \mu) = a^\top \Sigma_{12}\, b \;-\; \frac{\lambda}{2}(a^\top \Sigma_{11}\, a - 1) \;-\; \frac{\mu}{2}(b^\top \Sigma_{22}\, b - 1).$$

Stationarity w.r.t. a, b (i.e. differentiating w.r.t a, b and setting equal to zero) gives

$$\frac{\partial L}{\partial a}:\; \Sigma_{12} b - \lambda\, \Sigma_{11} a = 0, \qquad \frac{\partial L}{\partial b}:\; \Sigma_{21} a - \mu\, \Sigma_{22} b = 0,$$

and w.r.t. λ, μ recovers the constraints

$$a^\top \Sigma_{11} a = 1, \qquad b^\top \Sigma_{22} b = 1.$$

2. **Show that $\rho = \lambda$, $\rho = \mu$ and so that $\mu = \lambda$**

We have that

$$Cov(a^\top X, b^\top Y) = a^\top \Sigma_{12} b.$$

As $Var(a^\top X) = Var(b^\top Y) = 1$, the correlation is the same as the covariance, as so we have that

$$Corr(a^\top X, b^\top Y) = a^\top \Sigma_{12} b.$$

Pre-multiplying the first equation by $a^\top$ and the second by $b^\top$ yields

$$a^\top \Sigma_{12} b = \lambda, \qquad b^\top \Sigma_{21} a = \mu.$$

Accordingly $\rho = \lambda$ and $\rho = \mu$, and so $\rho = \lambda = \mu$.

3. **Solve for left-canonical variate loading vector**.

The KKT system becomes

$$\Sigma_{12} b = \rho\, \Sigma_{11} a, \qquad \Sigma_{21} a = \rho\, \Sigma_{22} b.$$

Solve the second for b

$$b = \frac{1}{\rho}\, \Sigma_{22}^{-1}\, \Sigma_{21} a,$$

and substitute into the first:

$$\Sigma_{12}\Big(\frac{1}{\rho}\Sigma_{22}^{-1}\Sigma_{21} a\Big) = \rho\, \Sigma_{11} a \quad \Longrightarrow \quad \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} a = \rho^2\, \Sigma_{11} a.$$

This is a generalized eigenproblem. We now create a regular eigenproblem, where the matrix involved is symmetric and positive definite.

Set

$$a = \Sigma_{11}^{-\frac{1}{2}}\, y.$$

Substituting the above in and moving all matrices to the left-hand side, we have that

$$\Sigma_{11}^{-\frac{1}{2}}\, \Sigma_{12}\, \Sigma_{22}^{-1}\, \Sigma_{21}\, \Sigma_{11}^{-\frac{1}{2}}\, y = \rho^2\, y,$$

i.e. $R\, y = \rho^2 y$ with $R = \Sigma_{11}^{-\frac{1}{2}}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\Sigma_{11}^{-\frac{1}{2}}$. As R can be written in the form LL', is is symmetric and positive (semi)-definite. Accordingly, the eigenvectors can be chosen orthogonal, and the eigenvalues are real and non-negative.

We choose the leading eigenpair (ρ^2, e_1), as the square root of the eigenvalue is the correlation, so we choose the largest eigenvalue.

From this, we recover

$$a_1 = \Sigma_{11}^{-\frac{1}{2}} e_1.$$

Now, we verify that $\mathrm{Var}(a_1^\top X) = 1$:

$$\mathrm{Var}(a_1^\top X) = \mathrm{Var}(e_1^\top \Sigma_{11}^{-\frac{1}{2}} X) = e_1^\top \Sigma_{11}^{-\frac{1}{2}} \mathrm{Var}(X) \Sigma_{11}^{-\frac{1}{2}} e_1 = e_1^\top \Sigma_{11}^{-\frac{1}{2}} \Sigma_{11} \Sigma_{11}^{-\frac{1}{2}} e_1 = e_1^\top I e_1 = 1.$$

4. **Solve for right-canonical variate loading vector**.

From differentiating the Lagrange function w.r.t. b, we have

$$\Sigma_{21} a = \rho \, \Sigma_{22} b.$$

which implies that

$$b_1 = \rho^{-1} \Sigma_{22}^{-1} \Sigma_{21} a_1$$

Substituting in $a_1 = \Sigma_{11}^{-\frac{1}{2}} e_1$, we can follow the same route as before to show that $\mathrm{Var}(b_1^\top Y) = 1$.

This equation is the basis for $b_1 \rho \Sigma_{22}^{-1} \Sigma_{21} a_1$ (as here the constant ρ^{-1} has just been dropped). So, one could work out $\mathrm{Var}((\Sigma_{22}^{-1} \Sigma_{21} a_1)' Y)$ and divide through by its square root to ensure unit variance, but you could also just divide by ρ.

5. **Solve for remaining canonical variate pair loadings**.

For subsequent canonical variates, we work with essentially the same Lagrangian function, except that there are extra constraint functions (orthogonality with earlier canonical variates).

We again find that $\rho = \lambda = \mu$, and find the same eigenvector equation. We are now restricted in our choice of eigenvectors to only choosing those orthogonal to previously-used eigenvectors, and so choose eigenvectors associated with the next largest eigenvalue. Verification of the unit-variance of a_i and the formula for b_i follow as before.

To satisfy the additional constraints of orthogonality, we first note that $a_i' \Sigma_{11} a_j = e_i^\top \Sigma_{11}^{-\frac{1}{2}} \Sigma_{11} \Sigma_{11}^{-\frac{1}{2}} e_j = e_i^\top e_j = 0$ for $i \neq j$. Similar reasoning holds for orthogonality of different right-canonical variates.

Finally, we note that $a_i' \Sigma_{12} b_j = a_i^\top \Sigma_{12} \rho^{-1} \Sigma_{22}^{-1} \Sigma_{21} a_j$. As the inner matrix $(\Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})$ is symmetric and $a_i \cdot a_j = 0$ for $i \neq j$, we have that this too is equal to zero, and the derivation is complete.